# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

A first order reaction is half-completed in 45 minutes. How long does it need for $99.9\%$ of the reaction to be completed?

**(A)** 20 hours

**(B)** 10 hours

**(C)** $7\frac{1}{2}$ hours

**(D)** 5 hours

**Q.2**

A reaction which is of first order w.r.t. reactant $\mathrm{A}$, has a rate constant $6\ \mathrm{min}^{-1}$. If we start with $[\mathrm{A}] = 0.5\ \mathrm{mol\ L}^{-1}$, when would [ $\mathrm{A}$ ] reach the value of $0.05\ \mathrm{mol\ L}^{-1}$

**(A)** $0.384\ \mathrm{min}$

**(B)** $0.15\ \mathrm{min}$

**(C)** $3\ \mathrm{min}$

**(D)** $3.84\ \mathrm{min}$

**Q.3**

The rate of a first order reaction is $1.5 \times 10^{-2}\ \mathrm{mol\ L}^{-1}\ \mathrm{min}^{-1}$ at $0.5\mathrm{M}$ concentration of the reactant. The halflife of the reaction is

**(A)** $0.383\ \mathrm{min}$

**(B)** $23.1\ \mathrm{min}$

**(C)** $8.73\ \mathrm{min}$

**(D)** $7.53\ \mathrm{min}$

**Q.4**

For a first order reaction $\mathrm{A} \longrightarrow \mathrm{B}$ the reaction rate at reactant concentration of $0.01\mathrm{M}$ is found to be $2.0 \times 10^{-5}\ \mathrm{mol\ L}^{-1}\ \mathrm{s}^{-1}$. The half life period of the reaction is

**(A)** $30\ \mathrm{s}$

**(B)** $220\ \mathrm{s}$

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**(C)** $300\ \text{s}$

**(D)** $347\ \text{s}$

**Q.5**

A reaction proceeds by first order, $75\%$ of this reaction was completed in 32 min. The time required for $50\%$ completion is

**(A)** $8\ \text{min}$

**(B)** $16\ \text{min}$

**(C)** $20\ \text{min}$

**(D)** $24\ \text{min}$

**Q.6**

In the reaction $\text{A} \rightarrow \text{B} + \text{C}$, rate constant is $0.001\ \text{Ms}^{-1}$. If we start with $1\text{M}$ of A then conc. of $\text{A}$ and B after 10 minuter are respectively.

**(A)** $0.5\text{M}, 0.5\text{M}$

**(B)** $0.6\text{M}, 0.4\text{M}$

**(C)** $0.4\text{M}, 0.6\text{M}$

**(D)** $0.6\text{M}0.5\text{M}$

**Q.7**

The half life for the virus inactivation if in the beginning $1.5\%$ of the virus is inactivated per minute is (Given: The reaction is of first order)

**(A)** $76\ \text{min}$

**(B)** $66\ \text{min}$

**(C)** $56\ \text{min}$

**(D)** $46\ \text{min}$

**Q.8**

A radioactive isotope having a half - life period of 3 days was received after 12 days. If $3\ \text{g}$ of the isotope is left in the container, what would be the initial mass of the isotope?

**(A)** $12\ \text{g}$

**(B)** $36\ \text{g}$

**(C)** $48\ \text{g}$

**(D)** $24\ \text{g}$

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**Q.9**

The rate constant of a reaction with a virus is $3.3 \times 10^{-4}\ \mathrm{s}^{-1}$. Time required for the virus to become $75\%$ inactivated is

**(A)** $35\ \mathrm{min}$

**(B)** $70\ \mathrm{min}$

**(C)** $105\ \mathrm{min}$

**(D)** $17.5\ \mathrm{min}$

**Q.10**

A Geigger Muller counter is used to study the radioactive process. In the absence of radioactive substance $A$, it counts 3 disintegration per second (dps). At the start in the presence of $\mathrm{A}$, it records 23 dps; and after 10 min 13 dps,
(i) What does it count after $20\ \mathrm{min}$
(ii) What is the halflife of A?

**(A)** $8\mathrm{dps}, 10\ \mathrm{min}$

**(B)** $5\mathrm{dps}, 10\ \mathrm{min}$

**(C)** $5\mathrm{dps}, 20\ \mathrm{min}$

**(D)** $5\mathrm{dps}, 5\ \mathrm{min}$

**Q.11**

The rate equation for a reaction,
$\mathrm{N_2O} \longrightarrow \mathrm{N_2} + 1/2\mathrm{O_2}$
is Rate $= \mathrm{k}\left[\mathrm{N_2O}\right]^0 = \mathrm{k}$. If the initial concentration of the reactant is $a$ mol $\mathrm{Lit}^{-1}$, the half-life period of the reaction is

**(A)** $t_{\frac{1}{2}} = \frac{a}{2k}$

**(B)** $-t_{\frac{1}{2}} = ka$

**(C)** $t_{\frac{1}{2}} = \frac{a}{k}$

**(D)** $t_{\frac{1}{2}} = \frac{k}{a}$

**Q.12**

Acid hydrolysis of ester is first-order reaction and rate constant is given by
$\mathrm{k} = \frac{2.303}{\mathrm{t}} \log \frac{\mathrm{V_\infty - V_0}}{\mathrm{V_\infty - V_t}}$
where $\mathrm{V_0},\ \mathrm{V_t}$ and $\mathrm{V}\infty$ are the volume of standard $\mathrm{NaOH}$ required to neutralize acid present at a given time; if ester is $50\%$ hydrolysed then

**(A)** $\mathrm{V_\infty} = \mathrm{V_t}$

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**(B)** $V_\infty = (V_t - V_0)$

**(C)** $V_\infty = 2\,V_t - V_0$

**(D)** $V_\infty = 2\,V_t + V_0$

**Q.13**

Half-lives of first-order and zeroth order reactions are same. Ratio of rates at the start of reaction is

**(A)** $0.693$

**(B)** $\frac{1}{0.693}$

**(C)** $2 \times 0.693$

**(D)** $\frac{2}{0693}$

**Q.14**

The plot of concentration of the reactant Vs time for a reaction is a straight line with a negative slope. The reaction follows a rate equation of

**(A)** zero order

**(B)** first order

**(C)** second order

**(D)** third order

**Q.15**

The reaction $L \longrightarrow M$ is started with $10.0$ g of L. After 30 and 90 minutes $5.0$ g and $1.25$ g of $L$ respectively are left. The order of the reaction is

**(A)** 0

**(B)** 1

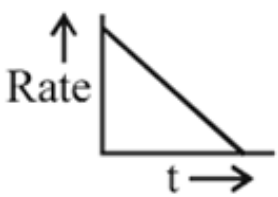
**(C)** 2

**(D)** 3

**Q.16**

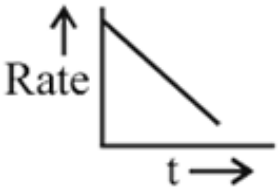
The plot that represents the zero order reaction is:

**(A)**

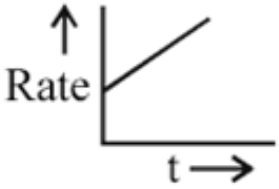



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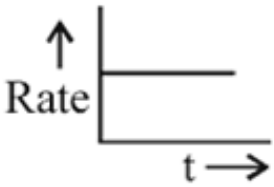
**(B)**




**(C)**




**(D)**




**Q.17**

For a first order reaction, a plot of $\log(a-x)$ against time is a straight line with a negative slope equal to

**(A)** $\frac{-k}{2.303}$

**(B)** $-2.303k$

**(C)** $\frac{2.303}{k}$

**(D)** $-\frac{E_a}{2.303R}$

**Q.18**

In the presence of an acid, the initial concentration of cane sugar was reduced from $0.20$ to $0.10\text{M}$ in 5 hours and from $0.2$ to $0.05\text{M}$ in 10 hours. The reaction is of :

**(A)** Zero order

**(B)** First order

**(C)** Second order

**(D)** Third order

**Q.19**

The half life period for catalytic decomposition of $AB_3$ at $50\ \text{mmHg}$ is $4\text{hrs}$ and at $100\ \text{mmHg}$ is $2\text{hrs}$. The order of reaction is

**(A)** 1

**(B)** 3

**(C)** 2

**(D)** 0

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**Q.20**

The hypothetical reaction $A_2 + B_2 \longrightarrow 2AB$; follows the following mechanism $A_2 \overset{\text{Fast}}{\longrightarrow} A + A, A + B_2 \overset{\text{Slow}}{\longrightarrow} AB + B, A + B \overset{\text{Fast}}{\longrightarrow} AB$ The order of the overall reaction is

**(A)** 0

**(B)** 1

**(C)** 2

**(D)** $3/2$

## Answers & Solutions

**Q.1 Answer:**

$7\frac{1}{2}$ hours

**Solution:**

$k = \frac{0.693}{45}$ **further**
$t = \frac{2.303}{0.693} \times 45 \log \frac{100}{100-99.9}$
$= \frac{2.303 \times 45 \times 3}{0.693} = 7\frac{1}{2}$ **hours.**

**Q.2 Answer:**

$0.384$ min

**Solution:**

$t = \frac{2.303}{k} \log \frac{a}{a-x}$
$= \frac{2.303}{6} \log \frac{0.5}{0.05} = 0.384$ min

**Q.3 Answer:**

$23.1$ min

**Solution:**

**For a first order reaction,** $A \rightarrow$ **Products**
$r = k[A]$ **or** $k = \frac{r}{[A]}$
$\Rightarrow k = \frac{1.5 \times 10^{-2}}{0.5} = 3 \times 10^{-2}$
**Further,** $t_{1/2} = \frac{0.693}{k} = \frac{0.693}{3 \times 10^{-2}} = 23.1$

**Q.4 Answer:**

$347$ s

**Solution:**

**Given** $[A] = 0.01M$
**Rate** $= 2.0 \times 10^{-5}$ mol $L^{-1}$ $s^{-1}$
**For a first order reaction Rate** $= k[A]$

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$k = \frac{2.0 \times 10^{-5}}{[0.01]} = 2 \times 10^{-3}$
$t_{1/2} = \frac{0.693}{2 \times 10^{-3}} = 347\text{sec}$

**Q.5 Answer:**

16 min

**Solution:**

**Given: 75% reaction gets completed in 32 min**
**Thus,** $k = \frac{2.303}{t} \log \frac{a}{(a-x)}$
$= \frac{2.303}{32} \log \frac{100}{(100-75)}$
$= \frac{2.303}{32} \log 4 = 0.0433 \text{ min}^{-1}$
**Now we can use this value of** $k$ **to get the value of time required for 50% completion of reaction**
$t = \frac{2.303}{k} \log \frac{a}{(a-x)} = \frac{2.303}{0.0433} \log \frac{100}{50}$
$= \frac{2.303}{0.0433} \log 2 = 16 \text{ min}$

**Q.6 Answer:**

0.4M, 0.6M

**Solution:**

$[\mathrm{A}]_{\mathrm{t}} = [\mathrm{A}] - \mathrm{kt} = 1 - 0.001 \times 10 \times 60$
$= 0.4\mathrm{M}$
$[\mathrm{B}]_{\mathrm{t}} = 0.001 \times 10 \times 60 = 0.6\mathrm{M}$

**Q.7 Answer:**

46 min

**Solution:**

**For the first order reaction for small finite change**
$k_1 = \frac{1}{[\mathrm{A}]} \frac{\Delta[\mathrm{A}]}{\Delta t} \Rightarrow \frac{\Delta[A]/[A]}{\Delta t} = 1.5\% \text{ min}^{-1}$
$= 0.015 \text{ min}^{-1}$
$t_{1/2} = \frac{0.693}{0015 \text{ min}^{-1}} = 46.2 \text{ min} \approx 46 \text{ min}$

**Q.8 Answer:**

48 g

**Solution:**

**Given** $t_{1/2} = 3$
**Total time** $\mathrm{T} = 12$
**No. of half lives** $(n) = \frac{12}{3} = 4$
$\left(\frac{1}{2}\right)^n = \frac{\mathrm{N}}{\mathrm{N_o}} \quad \therefore \left(\frac{1}{2}\right)^4 = \frac{3}{\mathrm{N}}$

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$\frac{3}{N} = \frac{1}{16}$
$\mathrm{N} = 48\ \mathrm{g}$

**Q.9 Answer:**

$70\ \mathrm{min}$

**Solution:**

$t_{1/2} = \frac{0.693}{k} = 2100\ \mathrm{s} = 35\ \mathrm{min}$
$t_{75\%} = 2t_{1/2} = 2 \times 35 = 70\ \mathrm{min}$

**Q.10 Answer:**

$8\mathrm{dps}, 10\ \mathrm{min}$

**Solution:**

**In the absence of $A$, 3 dps is zero error, hence**
**Initial count** $= 23 - 3 = 20\mathrm{dps}$
**After** $10\ \mathrm{min} = 13 - 3 = 10\mathrm{dps}$
**After** $20\ \mathrm{min} = 5\mathrm{dps}$
( **recorded** $= 5 + 3 = 8\mathrm{dps}$)
**(50\% fall in** $10\ \mathrm{min}, T_{50} = 10\ \mathrm{min}$)

**Q.11 Answer:**

$t_{\frac{1}{2}} = \frac{a}{2k}$

**Solution:**

**For a zero order reaction**
$t_{1/2} = \frac{a}{2k}$

**Q.12 Answer:**

$\mathrm{V}_\infty = 2\ \mathrm{V}_\mathrm{t} - \mathrm{V}_0$

**Solution:**

$$\mathrm{RCOOR'} + \mathrm{H_2O} \xrightarrow{H^+} \mathrm{RCOOH} + \mathrm{R'OH}$$

| | | |
|---|---|---|
| at t = 0 : a | 0 | 0 |
| at time t: (a − x) | x | x |
| at time ∞ : (a − a) | a | a |

**at** $\mathrm{t} = 0,\ \mathrm{V}_0 =$ **volume of** $\mathrm{NaOH}$ **due to** $\mathrm{H}^+$ **(catalyst)**
$\mathrm{V_t} = \mathrm{x} + \mathrm{V}_0$
$\mathrm{V}\infty = \mathrm{a} + \mathrm{V}_0$
**If ester is** $50\%$ **hydrolysed then** $x = \frac{a}{2}$
$V_t = \frac{a}{2} + V_0$

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$a = 2V_t - 2V_0$
$\therefore V_\infty = 2V_t - 2V_0 + V_0 = 2\ V_t - V_0$

**Q.13 Answer:**

$2 \times 0.693$

**Solution:**

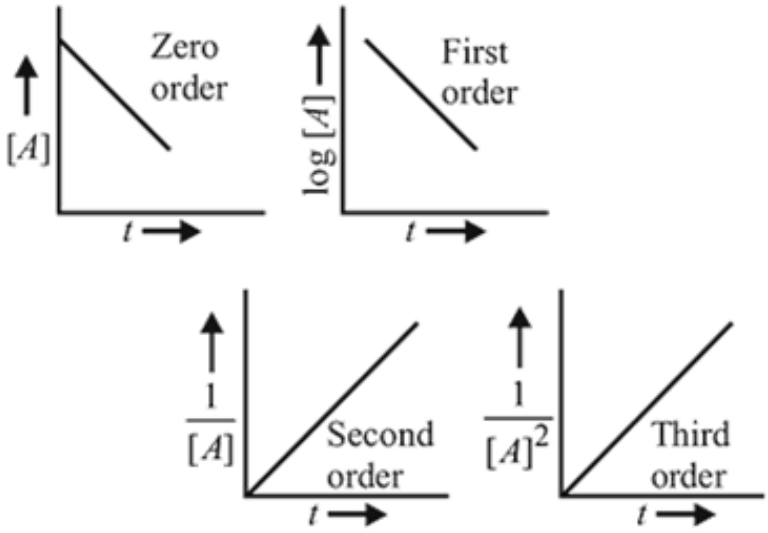
$T_{50}(\textbf{ zeroth }) = \frac{a}{2k_0}$
$T_{50}(\textbf{ first }) = \frac{0.693}{k_1}$
$\frac{a}{2k_0} = \frac{0.693}{k_1}$
$\therefore \frac{k_1}{k_0} = \frac{2\times 0.693}{a}$
$\frac{\left(\frac{dx}{dt}\right)_1}{\left(\frac{dx}{dt}\right)_0} = \frac{k_1 a}{k_0} = 2 \times 0.693$

**Q.14 Answer:**

zero order

**Solution:**

**Plots of conc. [A] Vs time, t**




**Q.15 Answer:**

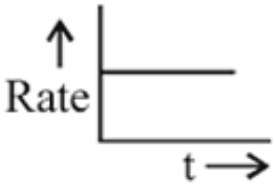
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**Solution:**

**After every 30 minutes the amount is reduced to $\frac{1}{2}$ therefore $t_{1/2}$ is 30 minutes. In 90 minutes the amount is reduced to $\frac{1}{8}$ i.e. $\frac{1}{2^n}$. Here $n = 3$. True for 1st order reaction.**

**Q.16 Answer:**

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**Solution:**

**For a zero order, rate of reaction does not change with time.**

**Q.17 Answer:**

$\frac{-k}{2.303}$

**Solution:**

$t = \frac{2.303}{k} \log \frac{a}{a-x}$

**or** $t = \frac{2.303}{k} \log a - \frac{2.303}{k} \log(a-x)$

$\log(a-x) = \frac{-k}{2.303} t + \log a$

**Q.18 Answer:**

First order

**Solution:**

$\mathrm{k_1} = \frac{1}{5} \mathrm{In} \frac{0.2}{0.1}; \mathrm{k_2} = \frac{1}{10} \mathrm{In} \frac{0.2}{0.05}; \mathrm{k_1} = \mathrm{k_2}$

**Q.19 Answer:**

2

**Solution:**

$t_{1/2} \propto \frac{1}{(p)^{n-1}}$ **where** $n$ **is the order of reaction**

$\frac{2}{4} = \left(\frac{50}{100}\right)^{n-1}$ **or** $\frac{1}{2} = \left(\frac{1}{2}\right)^{n-1}$

$n - 1 = 1; n = 2$

**Q.20 Answer:**

$3/2$

**Solution:**

$\mathrm{A_2 + B_2 \longrightarrow 2AB;}$

$\mathrm{A_2 \longrightarrow A + A}$ **(Fast);**

$\mathrm{A + B_2 \longrightarrow AB + B}$ **(Slow)**

**Rate law** $= \mathrm{k[A]\,[B_2]}$ **put value of** $\mathrm{[A]}$ **from Ist reaction since** $\mathrm{A}$ **is intermediate**

$\sqrt{\mathrm{k\,[A_2]}} = \mathrm{A}$

$\therefore$ **Rate law equation** $= \mathrm{K}\sqrt{\mathrm{k\,[A_2]}}\,\mathrm{[B_2]}$

$\therefore$ **Order** $= \frac{1}{2} + 1 = \frac{3}{2}$

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